

Uncertainty Aware Tree Search: Using LLM Internals to Allocate Exploration Budget

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Abstract

Large Language Models (LLMs) have demonstrated remarkable reasoning capabilities, often enhanced by techniques like Chain-of-Thought (CoT) prompting. However, these methods often rely on a fixed, and frequently excessive, computational budget, leading to inefficiency. Models may "overthink" simple problems or fail to explore sufficiently on complex ones. To address this, we introduce Uncertainty Aware Tree Search (UATS), a novel framework that dynamically allocates computational resources during reasoning by interpreting the model's internal states. UATS employs two lightweight probes trained on the LLM's internal activations: an *uncertainty probe* to guide exploration by predicting future semantic diversity and determining the branching factor of a search tree, and a *value probe* to guide exploitation by pruning less promising reasoning paths. This allows the search to adapt in real-time, broadening when the model is uncertain and deepening when it is confident. Our preliminary experiments show that the UATS exploration mechanism can effectively find correct reasoning paths, though overall performance is currently bottlenecked by the value probe. This work represents a step towards more adaptive and resource-efficient reasoning in LLMs.

1 Introduction

Large Language Models (LLMs) have become increasingly powerful, tackling complex reasoning tasks

across various domains. A key technique that has unlocked these capabilities is Chain-of-Thought (CoT) prompting [12], which encourages models to generate intermediate reasoning steps before arriving at a final answer. This process mimics human problem-solving and has been shown to significantly improve performance.

However, the success of CoT and its variants, such as self-consistency [13] and tree-based search methods, comes at a significant computational cost. These approaches often rely on generating multiple reasoning paths (i.e., Test-Time Scaling or TTS), consuming a large number of tokens. A critical limitation of current methods is their static allocation of computational budget, leading to two primary inefficiencies:

1. **Overthinking:** For simpler problems where the model is confident, generating multiple CoTs is redundant and wasteful [6].
2. **Under-exploring:** For complex problems, a fixed budget may not be sufficient to explore the reasoning space adequately, especially in the critical initial steps where errors are most costly [7].

This paper addresses the challenge of efficiently allocating computational budget. We propose **Uncertainty Aware Tree Search (UATS)**, a framework that dynamically adjusts its exploration-exploitation strategy based on the model's internal uncertainty. The core premise of UATS is that an LLM's internal hidden states contain decodable metacognitive signals about its own knowledge and uncertainty [1, 2].

Instead of relying on a fixed-width search, UATS uses lightweight probes to "listen to the model itself," guiding the construction of a reasoning tree. Specifically, an *uncertainty probe* predicts the semantic diversity of potential future answers, determining how many branches to explore at each step. A *value probe* then evaluates the promise of each branch, pruning those that are less likely to lead to a correct solution.

This adaptive approach allows the model to allocate more resources to uncertain steps (going "wider") and focus its efforts on promising paths when confident (going "deeper"). Our framework is designed to be computationally cheaper than standard Best-of-N sampling while providing greater interpretability into the model's reasoning process.

2 Related Work

The UATS framework is built upon three pillars of research: the fundamental discovery that LLM internal states are decodable, the development of methods to quantify uncertainty, and the design of adaptive architectures for reasoning.

2.1 Probing Internal States for Metacognitive Signals

The scientific justification for UATS rests on the finding that an LLM's hidden states encode information about its "cognitive" processes. Foundational work by [1] demonstrated that a simple classifier trained on LLM activations could reliably distinguish true statements from false ones, establishing that a representation of veracity exists internally. This was extended by [2], who showed that models possess a coherent sense of their own knowledge, which can be elicited via prompting to produce calibrated probabilities. These papers established the core principle that UATS relies on: that meaningful metacognitive signals can be extracted from hidden states using lightweight external modules.

2.2 Architectures for Adaptive Reasoning

UATS is part of a broader effort to design more dynamic reasoning architectures. Tree-based methods like Adaptive Branching MCTS (AB-MCTS) [5] also aim to balance the "wider" vs. "deeper" trade-off, but rely on an external reward signal, whereas UATS uses an internal signal from the model itself.

Another line of work focuses on **early exiting** to prevent overthinking. [6] uses a self-verification probe, conceptually identical to our value probe, to terminate reasoning once a high-confidence intermediate answer is found. UATS integrates this idea into a broader search, where the value probe's score is used for both pruning and contributing to the early-exit condition. The criticality of allocating budget to the **initial** reasoning steps has been empirically demonstrated by [7], who show that first-step errors are catastrophic. The dynamic branching of UATS is uniquely suited to address this by forcing a wide exploration at the beginning of the search where uncertainty is often highest.

2.3 Process Reward Models (PRMs) and the Value Probe

The UATS value probe is a form of Process Reward Model (PRM), a model trained to evaluate intermediate reasoning steps. Training effective PRMs is challenging. The common practice of using Monte Carlo (MC) rollouts to generate training labels is known to be noisy and unreliable [8]. This finding is critical, as it provides a direct explanation for the performance bottleneck we observed in our preliminary experiments. Recent work has focused on improving PRM calibration [9] and exploring more efficient training paradigms, such as deriving process rewards implicitly from cheaper outcome-level labels [10], pointing a clear path forward for improving the UATS value probe.

3 The Role of Uncertainty in UATS

The exploration mechanism in UATS is driven by a specific, nuanced form of uncertainty. It is not concerned with the model’s confidence in the next token, but rather with its uncertainty about the ultimate *meaning* of the final answer. This concept, termed “semantic uncertainty,” is critical for guiding a reasoning process effectively.

3.1 Defining and Measuring Uncertainty

A fundamental flaw in using simple token-level uncertainty (e.g., the entropy of the next-token probability distribution) is that language is redundant; many different wordings can convey the exact same idea. A model might be highly certain about an answer’s meaning (e.g., “Paris”) but express this through various lexically diverse phrases (“It’s Paris,” “The capital of France is Paris,” etc.), leading to a misleadingly high token-level entropy.

To solve this, [3] introduced **semantic entropy**, a robust metric that measures uncertainty at the level of meaning. The calculation is a multi-step, computationally intensive process:

1. **Sample:** Generate K distinct responses by sampling from the model’s output distribution.
2. **Cluster:** Group the K responses into clusters of semantic equivalence, typically using a separate Natural Language Inference (NLI) model.
3. **Compute Entropy:** Calculate the Shannon entropy over the distribution of these semantic cluster probabilities.

A high semantic entropy value indicates high semantic uncertainty and is a strong predictor of hallucinations, as it reveals that the model is considering multiple, semantically distinct possibilities. This metric provides the formal grounding for the signal our uncertainty probe aims to capture.

3.2 Approximating Uncertainty with Probes

While semantic entropy is a powerful diagnostic tool, its reliance on generating 5-10 full responses per query makes it prohibitively expensive for real-time applications like guiding a search tree. This computational barrier motivated a critical line of follow-up research focused on finding a more efficient way to access this signal.

The solution, developed in parallel to our work, is to distill the expensive, multi-sample metric into a cheap, single-pass probe. [4] introduced **Semantic Entropy Probes (SEPs)**, which are simple linear models trained to approximate the full semantic entropy score directly from the hidden state activations of a single generation. This “distillation” paradigm is precisely what the UATS uncertainty probe employs. We pre-compute the expensive semantic entropy scores for a large dataset offline, and then train a lightweight probe to predict this score from a single reasoning step’s internal state.

The crucial contribution of UATS is to move this concept from a passive, post-hoc filter to an active, prospective controller for a generative search algorithm. The probe’s output is not just a score, but a control signal that dynamically determines the branching factor of the search tree, thereby shaping the entire reasoning process.

3.3 The Value Probe and its Current Limitations

While the uncertainty probe governs exploration, the value probe is responsible for exploitation—identifying which of the explored paths are most promising. Our preliminary results strongly suggest that this component is the primary bottleneck in the UATS framework. One key challenge is the nature of the reasoning traces we are evaluating. Unlike many PRM experiments that operate on shorter, more structured reasoning, our CoTs can be very long and complex. This creates a difficult credit assignment problem for the value probe. A single early mistake can invalidate a long chain of otherwise correct reasoning, and conversely, a correct final

answer might be reached via a flawed path. Training a small MLP to capture these long-range dependencies from a single hidden state is a demanding task and likely a significant source of the probe’s current limitations.

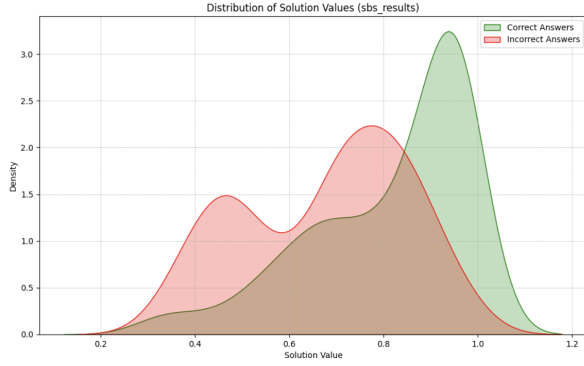


Figure 1: The current algorithm implementation is limited by the value probe. The value estimates for the distribution above are taken from the final beams of SBS. The overlap between the two distributions show that the probe is yet not able to clearly distinguish a proper and improper solution.

4 Uncertainty Aware Tree Search (UATS)

The UATS algorithm is designed to build a search tree whose structure adapts to the model’s real-time uncertainty. The core idea is to use two specialized probes that read the LLM’s internal state at each reasoning step to make decisions about branching (exploration) and pruning (exploitation).

4.1 Probes from Internal Activations

We utilize two small Multi-Layer Perceptrons (MLPs), referred to as probes, that take the hidden state activations of the last token of a reasoning step as input.

- **Uncertainty Probe (P_u):** This probe is trained to predict a score, u , that approximates

the semantic diversity of potential final answers reachable from the current reasoning state. A higher score indicates greater uncertainty and suggests that multiple distinct reasoning paths should be explored.

- **Value Probe (P_v):** This probe is trained as a Process Reward Model (PRM). It predicts a value, v , representing the probability that the current reasoning path will lead to a correct final answer. This score is used to rank and select the most promising branches.

4.2 Algorithm

The UATS algorithm proceeds as a guided tree search. Given a prompt, the model begins generating a CoT. The reasoning is segmented into steps (e.g., delimited by newlines). At the end of each step, the following process occurs:

1. **Uncertainty Estimation:** The hidden state of the final token of the current step is fed into the uncertainty probe P_u to get a score u . This score is then mapped to an integer, k_u , which determines the number of new branches to generate from the current node.
2. **Branching (Exploration):** The model generates k_u distinct continuations from the current reasoning step, creating k_u new child nodes in the tree.
3. **Pruning (Exploitation):** The value probe P_v is used to score all leaf nodes in the tree. To manage computational cost, we maintain only the top- W branches, where W is a maximum beam width hyperparameter. Less promising branches are pruned and not expanded further.
4. **Early Exit:** If the uncertainty probe consistently outputs a low score (e.g., u corresponds to $k_u = 1$) for a predefined number of consecutive steps along a single path, we consider the model to be "confident." The algorithm then performs an early exit, terminating the search and generating the final answer from the single most promising branch.

4.3 Probe Training

The probes are trained via supervised fine-tuning on data collected from the base LLM.

- **Value Probe Training:** We follow a methodology based on Monte Carlo (MC) rollouts. We generate CoTs, randomly cut them at various steps, and perform N completions from that point. A step is given a positive label if at least one of the rollouts reaches the correct final answer. The probe is trained via binary cross-entropy on this data. We acknowledge recent findings [8] that show this method produces noisy supervision and consider this a baseline approach.
- **Uncertainty Probe Training:** To generate labels for uncertainty, we adopt the paradigm of distilling a semantic entropy score [4]. We perform N short completions from a given reasoning step, forcing the model to output just the final answer. We then compute a semantic diversity metric over these N answers (e.g., entropy over semantic clusters [3]). The uncertainty probe is trained via regression to predict this score from the activations of the reasoning step’s final token.

This training paradigm allows us to replace an expensive inference-time calculation with a single, cheap forward pass of the probe.

5 Preliminary Experiments

We conducted a series of preliminary experiments to evaluate the feasibility and potential of UATS. All experiments were performed using the Qwen3-1.7B model on a 70-sample validation split from the MMLU-Pro dataset.

5.1 Experimental Setup

We compare UATS against several baselines:

- **Single Inference:** Standard greedy decoding with a single CoT generation.

- **Majority Voting:** Generating $n = 4$ independent CoTs and selecting the most frequent final answer.

- **Step-level Beam Search (SBS):** A fixed-width beam search with beam width $B1 = 4$.

For UATS, we set the maximum beam width to 4. We report two metrics: "Most consistent answer" (accuracy of the majority vote among final branches) and "At least one correct," which measures the potential of the search tree before the final pruning/selection step.

5.2 Results

Our initial results are summarized in Table 1. Note that these are from a single run with temperature sampling, so some variability is expected.

5.3 Discussion of Preliminary Results

These first results are a mixed but promising signal. The final accuracy of UATS, based on the most consistent answer from its top-ranked branches, does not yet surpass the baselines. However, the "at least one correct" metric tells a different story. This metric evaluates whether *any* branch generated within the entire UATS tree led to a correct answer, before the final selection by the value probe.

The UATS (all-branches) variant achieves a score of 68.57% on this metric, which is highly competitive. The significant gap between this score and the final accuracy (45.71%) is direct evidence of the value probe’s failure. It shows that the uncertainty-driven exploration is successfully generating correct reasoning paths, but the value probe is failing to identify and rank them correctly. This is consistent with our hypothesis that the long-chain credit assignment problem and noisy training data are severely limiting the value probe’s effectiveness.

Furthermore, we observe that for easy problems, UATS correctly generates a narrow, deep tree, conserving tokens and exiting early. This provides evidence that the uncertainty probe is capturing a meaningful signal for exploration.

Method	Most consistent (%)	At least one correct (%)
Single Inference	52.86	52.86
Majority Voting (n=4)	55.71	74.29
Majority Voting (n=4, budget=1024*4)	44.29	61.43
SBS (B1=4, B2=5)	57.14	-
UATS (beam=4, 8k, all-branches)	45.71	68.57

Table 1: Preliminary Results on MMLU-Pro (70 samples). ‘Most consistent’ refers to majority vote over final branches. ‘At least one correct’ measures if any branch in the final tree is correct. Unspecified methods are not budget-constrained.

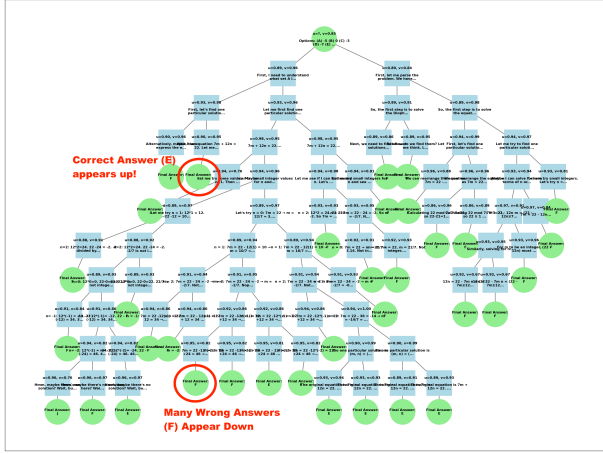


Figure 2: Example of a UATS tree for a math problem. The correct answer (E) appears relatively high in the tree, but the value probe fails to prioritize it, leading to extensive exploration of incorrect paths (F) below.

6 Future Work and Conclusion

This paper introduced Uncertainty Aware Tree Search (UATS), a framework for dynamically allocating computational budget during LLM reasoning. Our preliminary results indicate that the UATS framework has potential, but its performance is currently bottlenecked by the value probe.

Based on these findings, our future work will proceed in several key directions:

1. **Improved Value Probe Training:** The most critical next step is to replace the noisy MC rollout data generation. We will implement a

”consensus filtering” mechanism, integrating an LLM-as-a-judge with MC estimation to produce higher-quality labels, as proposed by [8]. We will also explore replacing the probe entirely with an implicit PRM derived from an Outcome Reward Model [10].

2. **Value Model vs. Value Head:** The current value probe is a small MLP. We will experiment with training a full value model (i.e., fine-tuning the entire LLM for the value prediction task) to see if a more powerful evaluator can overcome the current limitations.
3. **Scaling Experiments:** We will conduct more rigorous experiments on a larger validation set (200 samples) and analyze the performance of UATS across different computational budgets. A key hypothesis to test is whether UATS can push the performance plateau higher than methods like majority voting as the budget increases.
4. **Ablation Studies for Uncertainty Probe:** Inspired by [11], we will investigate whether a simpler linear probe, or even a fixed vector direction, could provide a sufficient signal for controlling exploration, as compared to our current non-linear MLP.

In conclusion, UATS presents a promising direction for building more efficient, adaptive, and interpretable reasoning systems. By listening to the LLM’s internal state of uncertainty, we can guide its computational effort more intelligently, paving the

way for models that can "think" more flexibly and effectively.

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